

ABHIJITH K J

+91 8129895738

abhijithkj3690@gmail.com

linkedin.com/in/abhijithkj3690

Portfolio

PROFESSIONAL SUMMARY

AI Engineer with 3 years of experience demonstrating the ability to design, build, secure, and operate end-to-end AI systems in production environments at **CDAC**. Proven expertise in developing LLM-powered agentic workflows, advanced RAG architectures, and multimodal systems integrating text and medical images. Adept at bridging LLMops practices with large-scale system design, collaborating with cross-functional teams to embed AI capabilities into software development and validation workflows under strict security and governance frameworks.

SKILLS

- GenAI & Agentic Systems:** Agentic AI, LangChain, LangGraph, MCP-based architectures, Tool Calling, Structured Outputs, Multi-Agent Systems, RAG, Chunking Strategies.
- LLMs & APIs:** OpenAI, Claude, Gemini, Llama 3, Prompt Engineering, AI Guardrails, Grounding, Hallucination Reduction, Vector Search, Embeddings (Qdrant).
- AI & Machine Learning:** PyTorch, TensorFlow, Recommendation Systems, XGBoost, Vision Transformers.
- Edge AI & Vision:** Model Quantization, NPU/GPU Optimization, YOLO, 3D U-Net, OpenCV.
- LLMOps & Architecture:** Python, FastAPI, Flask, Docker, Kubernetes, CI/CD, LLM Evaluation & Monitoring, Microservices-based architectures, Large-scale System Design.

PROFESSIONAL EXPERIENCE

Centre for Development of Advanced Computing (CDAC)

Mar 2024 – Present

AI Engineer | Thiruvananthapuram, Kerala

Generative AI, LLMs & Agentic Workflows

- Built **AI agents and orchestration pipelines** using **LangChain / LangGraph** and **MCP-based architectures**, embedding tool calling and structured outputs to automate complex reporting.
- Developed **multimodal AI systems integrating text and medical images** via BLIP-2 Q-Former, establishing robust AI pipelines that achieve 88% clinical domain accuracy.
- Implemented enterprise **RAG (Retrieval-Augmented Generation) pipelines** utilizing optimal chunking strategies, **prompt engineering**, **AI guardrails**, and **grounding** for hallucination reduction.

Foundation Models & Edge Architecture

- Directed **large-scale system design** and data pipelines for pre-training a 3D DINOv2 foundation model on massive, highly unstructured 3D volumetric datasets.
- Executed **model quantization and performance tuning** on deep learning models (nnU-Net), delivering a 97% Dice Score while slashing inference latency for real-time edge processing.
- Fine-tuned 303M-parameter foundation models, ensuring **efficient utilization of NPU / GPU resources** across distributed computing clusters (H100s).

LLMOps, Microservices & Distributed Systems

- Developed **microservices-based architectures** (FastAPI/Flask) for scalable AI deployment, integrating AI agents into distributed systems and service-oriented platforms utilized by 50+ end-users.
- Operated robust **LLMOps practices**, managing CI/CD, **LLM evaluation and monitoring**, and Federated Learning models under strict security and governance frameworks.
- Collaborated with cross-functional teams to embed AI capabilities into software development and validation workflows**, mentoring junior developers on production-grade ML practices.

PROJECTS

- Multimodal RAG Information Retrieval** (LangChain, Qdrant, FastAPI) — Architected a RAG pipeline utilizing **embeddings and vector search**; integrated local models and foundation APIs (**OpenAI, Claude, Gemini**) to generate citation-grounded answers.
- Audio-Visual Data Extraction LLM** (MedGemma, Whisper, PyTorch) — Built a multimodal pipeline for joint text, image, and speech transcription analysis, automating **90%+** of manual summarization tasks.
- Edge-Based Infrared Animal Detection** (YOLOv11, Twilio API, OpenCV) — Developed an edge-deployed computer vision prototype to analyze far-infrared railway feeds, implementing real-time tracking and automated API alerting to prevent wildlife collisions.
- Purkinje Gaze Tracking System** (OpenCV, Image Processing) — Engineered a high-precision gaze detection algorithm leveraging Purkinje image analysis to map eye movements on-screen.

EDUCATION

Kerala University of Digital Science Innovation and Technology, Thiruvananthapuram

Aug 2022 – Jun 2024

Master of Science in Computer Science (AI)

V.V. College, Calicut University, Palakkad

Jul 2019 – Apr 2022

Bachelor of Science in Computer Science

ACHIEVEMENTS & ACTIVITIES

- **Conference Organiser** — Led the organisation of an **International Brain-Computer Interface (BCI) Conference** hosted by **CDAC**, managing end-to-end event coordination. 2025
- **Vice-Captain** — South Zone Inter-University Football Tournament, SRM University, Chennai. 2023
- **Individual Champion & Medalist** — State and district athletic meets, Kerala. 2022